Harsh Vardhan Singh

ep23btech11012@iith.ac.in | harsh23.dbs@gmail.com | +91 9449578710 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.13
KENDRIYA VIDYALAYA NAL	Class XII (CBSE)	2023	94.40%
KENDRIYA VIDYALAYA NAL	Class X (CBSE)	2021	91.80%

WORK EXPERIENCE

Software Engineering Intern at Goldman Sachs

- Engineered an agentic harness for GSAM with specialized agents for internal dashboards, enabling tool-driven execution and automation of workflows across multiple subteams.
- Architected a hierarchical multi-agent system with subteam-specific agents coordinated by an overarching GSAM agent, enabling modular task decomposition and unified orchestration across financial workflows.

Software Engineering Intern at Sarvam AI

- Enhanced the debugging platform with node-level token cost telemetry across agent pipelines.
- Built a multi-turn evaluation suite for Sarvam agents on the MultiChallenge dataset across 273 conversations.
- Engineered agentic workflows for Indus, including Deep Research, Deep Search, Fast Search, and KB Search systems with adaptive retrieval, multi-step reasoning, and query-aware execution planning.

Software Engineering Intern at Prava AI

- Designed a graph based RAG retriever using ontologies, layout semantics, and hierarchical relationships for context aware document retrieval.
- Built a parallel VLM based document parser to extract bounding boxes and structural layouts**.** Reduced 10K-page PDF parsing time by 83% using heuristic and rule-based page stitching, achieving 98% structural accuracy.
- Developed a RAG and agent evaluation pipeline with LLM as a judge, along with statistical matching methods like BLEU/ROUGE and ontology aware retrieval over datasets like Google Frames and VidOR.

PROJECTS

MoE-GAT Transformer for Shower Disentanglement, Dr. Saranya Ghosh - Github

- Proposed a SOTA energy based MoE-GAT, reducing particle-splitting failures by over 90% compared to CNN baselines.
- Developed a Geant4-based CMS ECAL simulation pipeline with 100K generated events, leveraging Hungarian matching for set-based loss computation and Optuna for hyperparameter optimisation.

Agentic RAG System for Pathway (Inter-IIT Tech Competition - 2nd Place) - Drive

- Designed an agentic pipeline with hybrid retrieval, secure doc parsing, vector stores, and LlamaGuard guardrails.
- Built a multi-agent RAG system with web search, code execution, query reformulation, and memory via LSA, achieving 94% relevance on legal/financial QA.

PINN Solver for Schrödinger's Equation - Github

- Trained a PINN to solve the 1D TISE by enforcing PDE residuals, boundary conditions, normalization, and energy deviation.
- Extended the model to learn unknown energy eigenvalues via joint optimization and curriculum learning, achieving high fidelity without mesh-based solvers.

Chaotic XOR-Stream Encryption for Multimedia - Github

- Built a logistic map based encryption system for real time image, audio, and video security across diverse file formats.

- Achieved lossless, reversible decryption with FFmpeg based media handling and chaos sensitive key stream generation.

SCHOLASTIC ACHIEVEMENTS

- Secured **2nd place** out of 23 IITs at **Inter-IIT Tech Meet 13.0**, India's premier student-led tech championship.
- Recipient of a scholarship from **Ministry Of Science and Technology** for academic excellence.

SKILLS

Languages: C, C++, Python, JavaScript, SQL, Bash.

Tools & Technologies: Git, Docker, Weights & Biases, TensorBoard, LaTeX, Postman, Shell Scripting.

Libraries & Frameworks: PyTorch, DDP, FSDP, TensorFlow, Scikit-learn, XGBoost, OpenCV, Transformers, Keras, MLflow.

POSITIONS OF RESPONSIBILITY

Overall Head at Epoch, IIT Hyderabad

- Leading a 30-member core team to drive AI/ML initiatives at IITH, organizing workshops, speaker sessions, and research sprints for the campus community.

Core Member at Tinkerers' Lab, IIT Hyderabad

- Collaborated with an 9-member ML team to explore and build AI projects, including GAN-based generation and YOLO-based object detection.

Teaching Assistant - Electricity and Magnetism || Special Relativity

RELEVANT COURSES

Computer Science: Data Structures and Applications, Computational Physics, Machine Learning for Physical Science, Data Science Analysis, introduction to Artificial Intelligence, Introduction to Programming.

Mathematics: Probability and Random Variable, Linear Algebra, Statistics, Advanced Mathamatics for Physics, Non-linear Dynamics, Differential Equations, Linear Systems and Signal Processing.